# PROPRANOLOL

Contraindicated in asthma

Masks hypoglycaemia

**CHEMICAL NAME** (RS)-1-(isopropylamino)-3-(naphthalen-1-yloxy)propan-2-ol (hydrochloride)
**CLASS** Non-selective β-adrenoceptor blocker (lipophilic)
**TRADE NAMES** Inderal · Inderal LA · Hemangeol (infants) · Propranolol (generic)

## THERAPEUTIC USES / INDICATIONS

- Hypertension (not first-line); angina; arrhythmias.
- Prophylaxis of migraine; essential tremor.
- Anxiety (somatic symptoms), thyrotoxicosis (symptom control).
- Portal hypertension (variceal bleeding prophylaxis); infantile haemangioma.

## ADMINISTRATION

- **PO** with or after food (↑ bioavailability); MR capsules once daily.
- **IV** slowly (1 mg/min) with ECG monitoring — rarely.
- Do not stop abruptly — taper over 1–2 weeks.

## DOSAGES

- **Hypertension:** 80 mg twice daily (max 160–320 mg/day).
- **Migraine/tremor:** 40 mg 2–3 times daily (or 80–160 mg LA daily).
- **Anxiety:** 40 mg daily, up to three times daily.
- **Thyrotoxicosis:** 10–40 mg 3–4 times daily.
- **Portal hypertension:** 40 mg twice daily, titrate to HR ↓ 25%.

## PHARMACOKINETICS

- **Onset:** PO 1–2 h. **Duration:** 6–12 h (LA 24 h).
- Extensive first-pass metabolism (bioavailability ~25%); crosses blood–brain barrier.
- Metabolised in liver; excreted in urine. **t½:** 3–6 h.

## NURSING MANAGEMENT

- Check apical pulse and BP before each dose; hold if HR <50–60 or SBP <90 (per order).
- Ask about asthma/wheeze.
- Monitor glucose closely in diabetics.
- Monitor for signs of heart failure.
- Taper; never stop suddenly.

## PHARMACODYNAMICS

- Blocks **β1** (heart: ↓ HR, contractility, renin) and **β2** (bronchi, vessels, liver glycogenolysis) receptors.
- Membrane-stabilising action; ↓ peripheral T4→T3 conversion.

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Fatigue, cold hands and feet.
- Bradycardia, dizziness.
- Sleep disturbance, nightmares, depression.

### ADVERSE EFFECTS (SERIOUS)

- Bronchospasm (asthma, COPD).
- Heart block, heart failure exacerbation.
- Masks signs of hypoglycaemia (tremor, tachycardia); delays recovery.
- Rebound angina/MI on sudden withdrawal.

## CONTRAINDICATIONS & CAUTIONS

- Asthma or history of bronchospasm.
- 2nd/3rd-degree heart block, sick sinus syndrome, severe bradycardia.
- Cardiogenic shock, uncontrolled heart failure.
- Prinzmetal angina; severe peripheral arterial disease; metabolic acidosis.

### KEY INTERACTIONS

- **Verapamil, diltiazem (IV):** asystole, heart block.
- **Insulin, sulfonylureas:** masked hypoglycaemia.
- **Clonidine:** rebound hypertension on withdrawal.
- **NSAIDs:** ↓ antihypertensive effect.

## PATIENT EDUCATION

- Do not stop suddenly.
- Diabetics: hypoglycaemia may show as sweating only — check glucose.
- Report wheezing, breathlessness, slow pulse or swelling.
- May cause tiredness and cold hands.

## KEY POINTS

- Non-selective β-blocker; crosses BBB.
- Never in asthma.
- Migraine, tremor, anxiety, thyrotoxicosis, varices.
- Masks hypoglycaemia; taper to stop.

**SOURCES CHECKED** **[294]** Propranolol 10 mg film-coated tablets (UK SmPC) **[295]** Propranolol, IV (US label)

*Full references p. 432–457*